

CUSTOMER CHURN ANALYSIS & RETENTION STRATEGY

End-to-End Banking Analytics — Full Project Report

Domain: Banking Analytics

Version 2.0 — Final

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Dataset: Bank Customer Churn · 10,000 records · 2016–2019

Stack: Python · SQL · Snowflake · Power BI · Jira

Part I reproduces the Business Requirement Document (v1.1) and the full build record — requirements, data engineering, star-schema modelling, exploratory analysis, and SQL business questions. Part II records the delivered Power BI dashboard, finalises the KPI baseline, and closes the project.

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Part I — Project Definition & Build

This part reproduces the Business Requirement Document (v1.1) and the full build record: requirements, data engineering, modelling, exploratory analysis, and SQL business questions. Part II covers dashboard delivery and closes the project.

1. Project Overview

This project analyses bank customer data to identify the key factors influencing customer churn and to provide data-driven retention strategies. The analysis supports management in understanding customer behaviour, churn patterns, and opportunities for improving engagement.

Dataset summary

- Demographics — age, gender, geography
- Financial indicators — balance, credit score, estimated salary
- Banking behaviours — tenure, activity status, credit card ownership
- Customer status — exited or retained

Attribute	Detail
Source	Local folder (Bank_Churn.csv)
Rows	10,001 rows in source (10,000 records + header)
Type	Structured tabular data
Destination (raw layer)	Snowflake — schema RAW.BANK_CHURN

2. Business Objectives

- Identify characteristics of customers likely to churn.
- Quantify churn rate and segment customers by risk level.
- Develop actionable insights for retention strategy.
- Enable management to monitor churn KPIs through Power BI dashboards.

3. Business Questions

- What is the overall churn rate across all customers?
- How do age, tenure, and balance influence churn likelihood?
- Are inactive customers or those without credit cards more likely to churn?
- Which geography or gender shows higher churn trends?
- What common patterns exist among high-balance but low-activity customers?
- How do credit score segments correlate with churn?
- What is the retention trend by tenure segment?

4. Data Gathering

Source: Bank_Churn.csv — customer churn dataset including demographics, credit, and account information. 10,001 rows (10,000 data rows plus header).

Column	Description	Type	Business use
RowNumber	Record index	Integer	Ignored in analysis
CustomerId	Unique customer ID	Integer	Identifier
CreditScore	Credit rating	Numeric	Creditworthiness measure
GeographyId	1=France, 2=Spain, 3=Germany	Categorical	Region segmentation
GenderId	1=Male, 2=Female	Categorical	Gender-based insights
Age	Customer age	Numeric	Key predictor for churn
Tenure	Years with bank	Numeric	Customer loyalty
Balance	Account balance	Numeric	Financial indicator
NumOfProducts	Products owned	Numeric	Engagement measure
HasCrCard	1=Yes, 0=No	Binary	Product ownership
IsActiveMember	1=Yes, 0=No	Binary	Engagement level
EstimatedSalary	Estimated annual income	Numeric	Income category
Exited	1=Yes, 0=No	Binary	Churn indicator
Bank DOJ	Date joined bank	Date	Tenure validation

5. Data Dictionary

- **RowNumber** — record number; no effect on output.
- **CustomerId** — random values; no effect on churn.
- **Surname** — no impact on the decision to leave.
- **CreditScore** — can affect churn; higher scores generally less likely to leave.
- **Geography** — location can affect the decision to leave.
- **Gender** — worth exploring whether gender plays a role.
- **Age** — older customers were expected to be less likely to leave (the data contradicts this; see Section 11).
- **Tenure** — years as a client; longer-tenured clients tend to be more loyal.
- **Balance** — customers with higher balances were expected to churn less.
- **NumOfProducts** — number of products purchased through the bank.

- **HasCrCard** — cardholders tend to churn less.
- **IsActiveMember** — active customers are less likely to churn.
- **EstimatedSalary** — higher earners were expected to churn less.
- **Exited** — whether the customer left (1 = Exit, 0 = Retain).
- **Bank DOJ** — date the customer joined the bank.

Credit score bands

Band	Range
Excellent	800 – 850
Very Good	740 – 799
Good	670 – 739
Fair	580 – 669
Poor	300 – 579

6. Business Rules

Rule	Description
Churn definition	Customers with Exited = 1 are churned.
Activity	IsActiveMember = 0 → low engagement risk group.
Location mapping	1 = France, 2 = Spain, 3 = Germany
Gender mapping	1 = Male, 2 = Female
Data validation	Exclude rows with missing or invalid CustomerId.
Date handling	Convert Bank DOJ to datetime and validate Tenure.

7. Deliverables

Deliverable	Description	Owner	Epic
BRD	BRD & FRD	BA	Epic 1
Snowflake database setup	Create BANK_CHURN DB and schemas	Data Engineer	Epic 2
Cleaning & transformation	Python/SQL preprocessing scripts	Data Engineer	Epic 2
Star-schema data model	Fact_CustomerChurn + dimensions	Data Modeler	Epic 3
EDA report	Insights and trend analysis	Data Analyst	Epic 3
Power BI dashboard	Interactive churn dashboard	BI Developer	Epic 4
Row-level security	Restrict views by geography	BI Admin	Epic 5
Data refresh pipeline	Scheduled refresh via gateway	BI Admin	Epic 5
Documentation package	ER diagram, dictionary, KPIs	BA	Epic 6

8. Stakeholders & Responsibilities

This project was delivered single-handedly; each role below was performed by the same person, which is why every contact is identical.

Role	Responsibility	Contact
Business Analyst	Requirements, BRD/FRD, EDA	Nasrul
Data Engineer	Snowflake warehouse, pipelines	Nasrul
Data Analyst	Churn analysis, KPI calculation	Nasrul
Data Modeller	Analytical star schema design	Nasrul
Power BI Developer	Dashboards, RLS, refresh	Nasrul
End Users	Consume dashboards, feedback	Nasrul

9. KPIs & Success Metrics

Note: the original targets below were set before analysis. The churn target was subsequently revised to 16.5% on evidence grounds — see Part II, Section 1.2.

KPI	Formula	Original target
Customer churn rate	$(\text{Churned} / \text{Total}) \times 100$	$\leq 10\%$
Customer retention rate	$100 - \text{churn rate}$	$\geq 90\%$
Active member ratio	$(\text{Active} / \text{Total}) \times 100$	$\geq 70\%$
Avg credit score by group	$\text{AVG}(\text{CreditScore})$	Retained > churned
Avg balance by group	$\text{AVG}(\text{Balance})$	Retained > churned
High-risk segment %	$(\text{Predicted} / \text{Total}) \times 100$	Monitor trend
Dashboard adoption rate	$(\text{Active users} / \text{Total}) \times 100$	$\geq 80\%$
Refresh success rate	$(\text{Success} / \text{Total}) \times 100$	100%

10. Dependencies & Constraints

Dependency	Description
Data availability	Dataset must be loaded to Snowflake before transformation.
Tool access	Power BI Pro licence and Snowflake credentials required.
Security	RLS must align with internal access policy.
Environment	Stable gateway for automated refreshes.

11. Data Flow

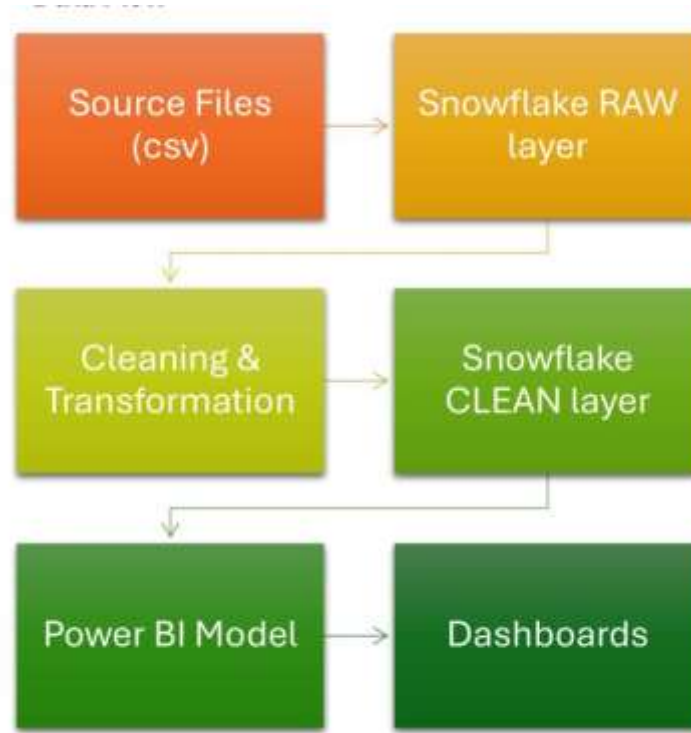


Figure 1 — Source-to-dashboard data flow

Refresh frequency

Source	Owner	Frequency
bank_churn.csv	Data Team	Weekly
Snowflake RAW schema	Data Engineering	Daily
Snowflake CLEAN schema	Data Engineering	Daily
Power BI dataset	BI Team	Daily / on refresh

KPI mapping to data fields

KPI	Calculation	Related fields
Churn rate	(#Exited / Total customers)	Exited
Average credit score	AVG(CreditScore)	CreditScore
Avg tenure	AVG(Tenure)	Tenure
Active vs inactive	COUNT(IsActiveMember)	IsActiveMember
Product penetration	AVG(NumOfProducts)	NumOfProducts
Avg balance by geography	AVG(Balance) GROUP BY GeographyID	Balance, GeographyID

12. Data Engineering & Warehouse Setup

12.1 Warehouse & schema setup

As a data engineer, I want to set up the data warehouse to host raw and clean data layers so that data is organised and ready.

Name	Purpose	Description
BANK_WH	Project warehouse	Warehouse for the churn analysis project
BANK_ANALYTICS_DB	Project database	Database for churn analysis and retention
...DB.RAW	Raw schema	Stores raw uploaded datasets, unmodified
...DB.CLEAN	Cleaned schema	Stores cleaned and transformed datasets
...DB.ANALYTICS	Analytics schema	Models and views consumed by Power BI

12.2 Load source data (RAW layer)

As a data engineer, I want to load source data into the RAW layer so that it is available for cleaning and analysis.

```
SELECT * FROM RAW.BANK_CHURN LIMIT 10;
SELECT COUNT(*) AS total_rows FROM RAW.BANK_CHURN; -- expect 10,000
```

Check	Expected	Actual	Status
Row count	10,000 data rows	10,000 (file has 10,001 incl. header)	Passed
Column headers	14 columns	14 columns	Passed
Data type consistency	Matches CSV	Matches CSV	Passed
Null values	0 in key fields	Verification pending	Action required

12.3 Data cleaning & transformation

As a data engineer, I want to extract, clean, and transform the data so that it is accurate, standardised, and ready for modelling.

a. Extract RAW data from Snowflake

```
import os
import pandas as pd
from snowflake.connector import connect

details = {
    'user': os.environ['SNOWFLAKE_USER'],
    'password': os.environ['SNOWFLAKE_PASSWORD'],
    'account': os.environ['SNOWFLAKE_ACCOUNT'],
    'warehouse': 'BANK_WH',
    'database': 'BANK_ANALYTICS_DB',
    'schema': 'RAW'
}
conn = connect(**details)
df = pd.read_sql('SELECT * FROM BANK_CHURN', conn)
```

Security note: credentials load from environment variables (or a git-ignored .env file) so they never enter source control. If this project history was ever pushed publicly with credentials inline, rotate the password and audit the account for unexpected activity.

b. Clean & transform

```
df = df.rename(columns={
    'ROWNUMBER': 'row_number', 'CUSTOMERID': 'customer_id',
    'CREDITSCORE': 'credit_score', 'GEOGRAPHYID': 'geography_id',
    'GENDERID': 'gender_id', 'AGE': 'age', 'TENURE': 'tenure',
    'BALANCE': 'balance', 'NUMOFPRODUCTS': 'num_of_products',
    'HASCRCARD': 'has_cr_card', 'ISACTIVEMEMBER': 'is_active_member',
    'ESTIMATEDSALARY': 'estimated_salary', 'EXITED': 'exited',
    'BANK_DOJ': 'bank_doj'
})

df['bank_doj'] = pd.to_datetime(df['bank_doj'])
df['geography_name'] = df['geography_id'].map({1: 'France', 2: 'Spain', 3: 'Germany'})
df['gender'] = df['gender_id'].map({1: 'Male', 2: 'Female'})
df['retain_status'] = df['exited'].map({1: 'Exit', 0: 'Retain'})
df = df.drop(columns=['row_number'])
```

c. Load cleaned data to the CLEAN layer

```
from snowflake.connector.pandas_tools import write_pandas

success, nchunks, nrows, _ = write_pandas(
    conn=conn, df=df, table_name='BANK_CHURN_CLEANED',
    schema='CLEAN', auto_create_table=True, overwrite=True
)
```

Check	Expected	Status
Row count matches RAW	10,000	Passed
Null values cleaned	0 in key columns	Passed — pending reverification
Category mapping	France/Spain/Germany; Male/Female	Passed
Business logic	Churn correctly derived	Passed

13. Data Modelling & Exploration

13.1 Star schema design

As a data modeler, I want to design a star schema so that Power BI and analysts can efficiently query, aggregate, and visualise customer churn insights.

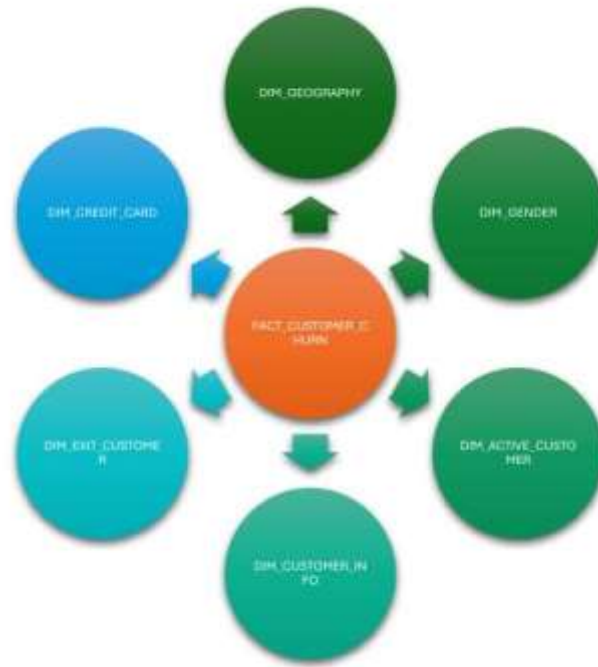


Figure 2 — Star schema: one fact table, six dimensions

FACT_CUSTOMER_CHURN_V1

Column	Type	Description
customer_id	INT	FK to DIM_CUSTOMER
geography_id	INT	FK to DIM_GEOGRAPHY
gender_id	INT	FK to DIM_GENDER
exit_id	INT	FK to DIM_EXIT_CUSTOMER
active_id	INT	FK to DIM_ACTIVE_CUSTOMER
cr_card_id	INT	FK to DIM_CREDIT_CARD
credit_score	INT	Customer credit score
age	INT	Customer age
tenure	INT	Years with bank
balance	FLOAT	Account balance
num_of_products	INT	Products owned
estimated_salary	FLOAT	Annual income

Column	Type	Description
churned	VARCHAR	Exited / Retained
bank_doj	TIMESTAMP	Load timestamp

Dimension tables: DIM_CUSTOMER (customer_id PK, customer_name), DIM_GEOGRAPHY (geography_id PK, country_name), DIM_GENDER (gender_id PK, gender), DIM_EXIT_CUSTOMER (exit_id PK, exit_category), DIM_ACTIVE_CUSTOMER (active_id PK, active_category), DIM_CREDIT_CARD (credit_id PK, category).

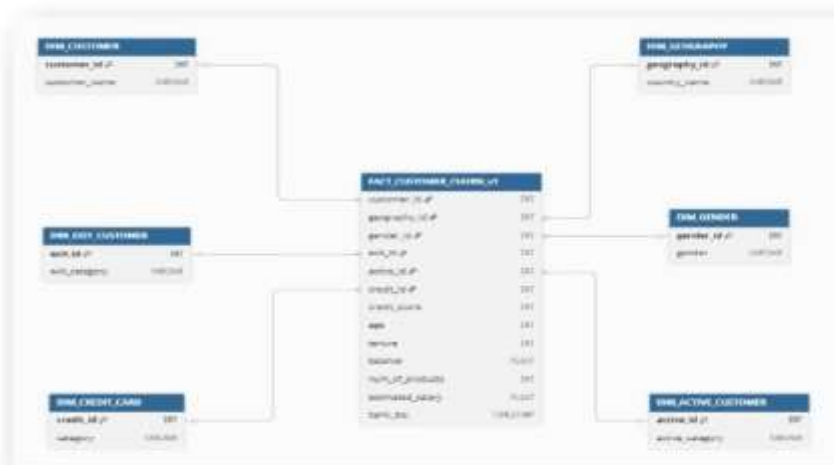


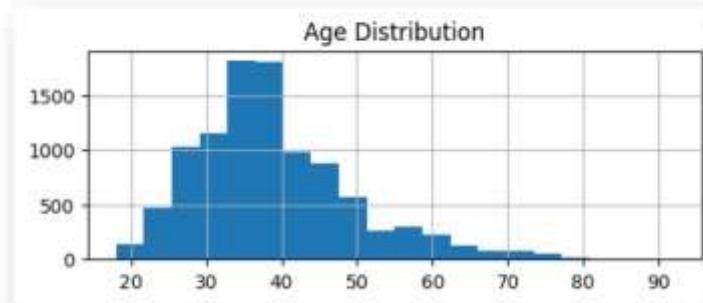
Figure 3 — ER diagram

14. Exploratory Data Analysis

As a data analyst, I want to perform exploratory analysis on the cleaned data so that I can understand patterns, relationships, and factors contributing to customer churn.

14.1 Univariate analysis

a) a. Age Distribution



* Majority of customers are aged 35-40 years old

Figure 4 — Age distribution: the majority of customers are aged 35–40.

b) Credit Score Distribution



* Credit Scores cluster around 600-700 range

Figure 5 — Credit score distribution: scores cluster in the 600–700 range.

c) Churned Distribution

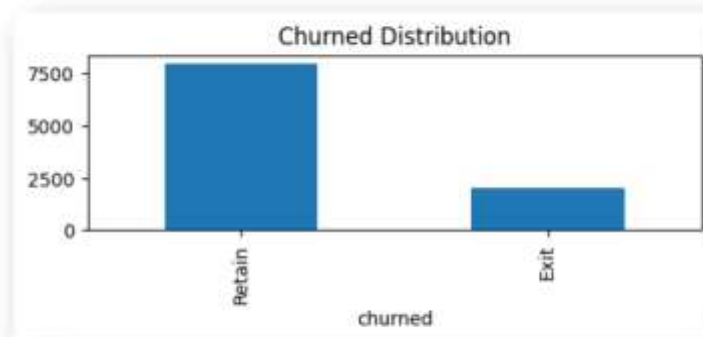


Figure 6 — Churned distribution: roughly 8,000 retained against 2,000 exited.

14.2 Bivariate analysis

Churn By Geography

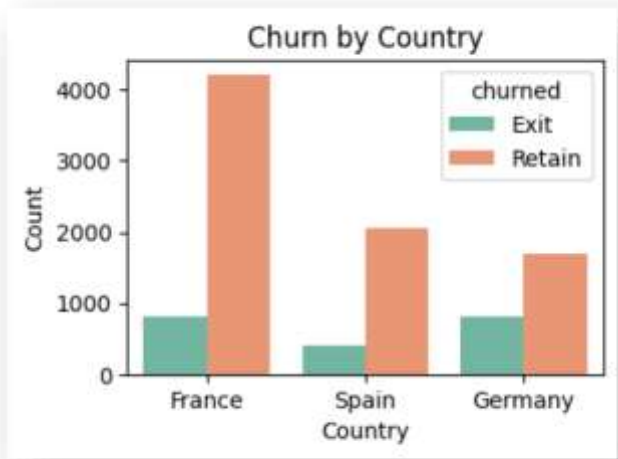


Figure 7 — Churn by geography

Churn By Gender

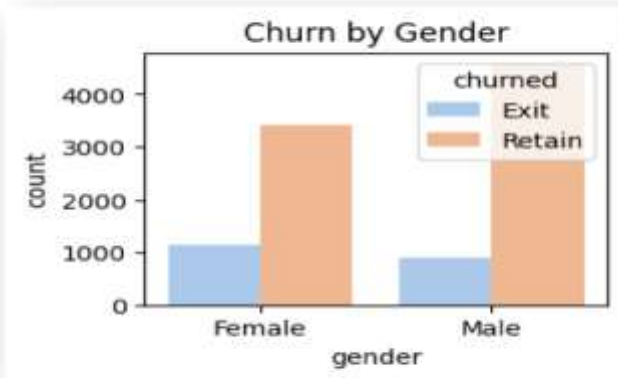


Figure 8 — Churn by gender

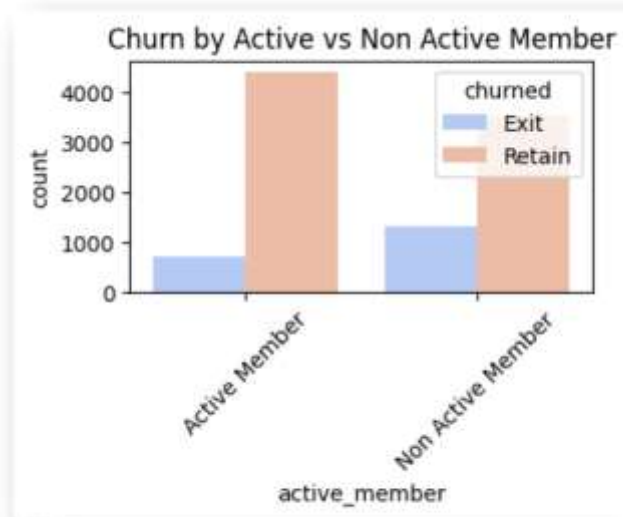


Figure 9 — Churn by active member status

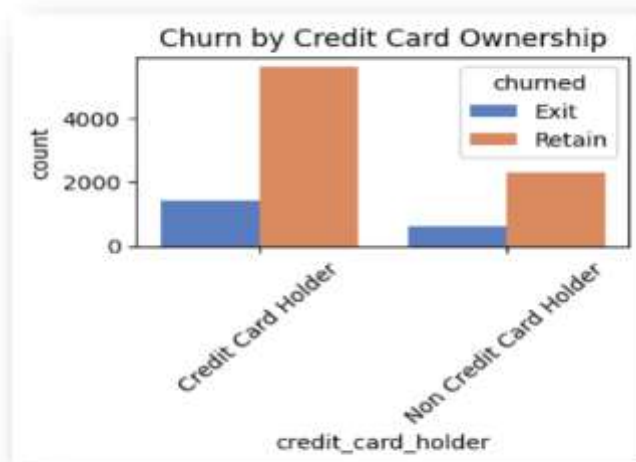


Figure 10 — Churn by credit card ownership

14.3 Correlation matrix

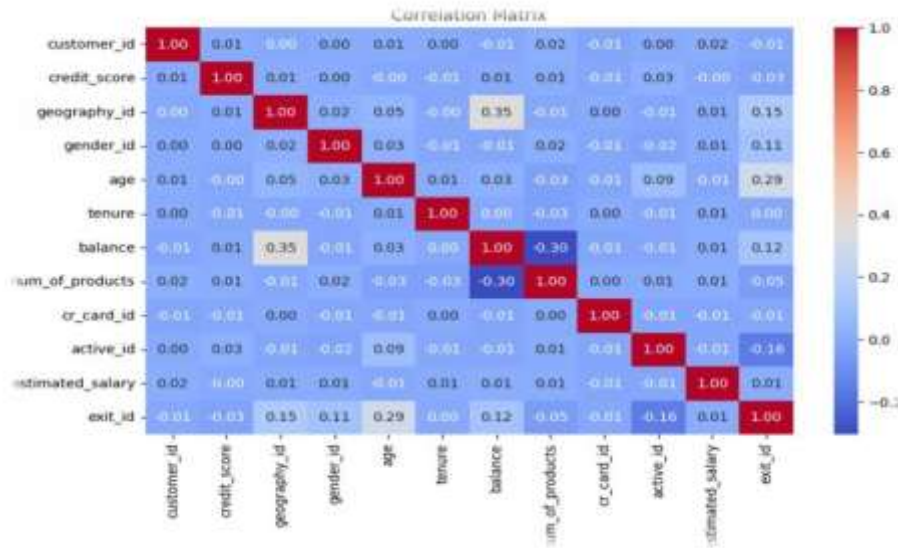


Figure 11 — Correlation matrix across all numeric fields

Metric	Score	Interpretation
Age	0.29	Weak-to-moderate positive — strongest single numeric correlation, but still modest.
Geography	0.15	Weak; confirm via segmentation rather than correlation alone.
Balance	0.12	Weak; higher balances show slightly higher churn tendency.
Gender	0.11	Weak; treat as a hypothesis to test, not a confirmed effect.
Active member	-0.16	Weak-to-moderate negative — active members churn less.
Estimated salary	0.01	Negligible.
Tenure	0.001	Negligible — no linear relationship in this dataset.
Credit card ownership	-0.01	Negligible.
Credit score	-0.03	Negligible.
Number of products	-0.05	Negligible-to-weak.

14.4 Segmentation analysis

Country	Total customers	Churned	Churn rate %
Germany	2,509	814	32.44
Spain	2,477	413	16.67
France	5,014	810	16.15

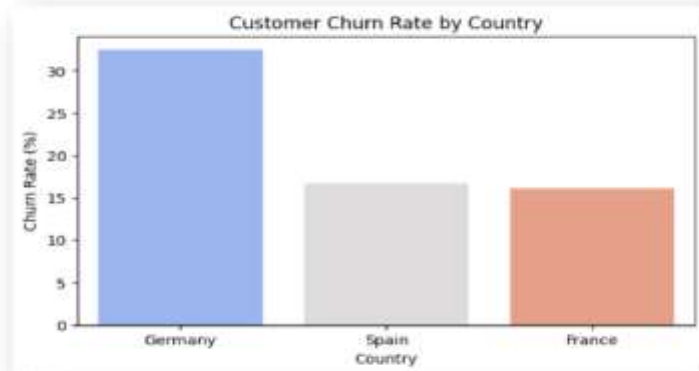


Figure 12 — Customer churn rate by country

- Germany shows the highest churn rate (32.4%), almost double France and Spain.
- France has the largest customer base (5,014) yet maintains the lowest churn rate (16.15%).
- Spain sits in between — moderate churn (16.7%) with a slightly smaller base than France.

15. Core Business Questions (SQL)

As a data analyst, I want to answer core business questions using SQL so that I can validate business KPIs using the cleaned data.

15.1 Overall churn rate

```
SELECT  
  ROUND(SUM("exit_id") * 100 / COUNT(*), 2) AS churn_rate_percentage  
FROM BANK_CHURN_CLEANED;
```

Result: 20.37%

15.2 Churn rate by country

```
SELECT  
  "country_name",  
  ROUND(SUM("exit_id") * 100 / COUNT(*), 2) AS churn_rate_percentage  
FROM BANK_CHURN_CLEANED  
GROUP BY ALL;
```

Country	Churn rate %
France	16.15
Spain	16.67
Germany	32.44

15.3 Churn rate by gender

Gender	Churn rate %
Female	25.07
Male	16.46

15.4 Churn rate by age group

```
SELECT
CASE
  WHEN "age" < 30 THEN 'Under 30'
  WHEN "age" BETWEEN 30 AND 39 THEN '30s'
  WHEN "age" BETWEEN 40 AND 49 THEN '40s'
  WHEN "age" BETWEEN 50 AND 59 THEN '50s'
  ELSE '60s+'
END AS age_group,
ROUND(SUM("exit_id") * 100 / COUNT(*), 2) AS churn_rate_percentage
FROM BANK_CHURN_CLEANED
GROUP BY ALL ORDER BY churn_rate_percentage DESC;
```

Age group	Churn rate %
50s	56.04
40s	30.79
60s+	27.95
30s	10.88
Under 30	7.56

15.5 Churn rate by product holding

Products held	Churn rate %
1	27.71
2	7.58
3	82.71
4	100.00

15.6 Average credit score and balance

Churn status	Avg credit score	Avg balance
Exit	645.35	91,108.54
Retain	651.85	72,745.30

15.7 Average tenure and age

Churn status	Avg tenure	Avg age
Retain	4.86	37
Exit	4.87	45

15.8 Summary of findings

KPI	Insight
Churn rate	20.37% of customers churned.
Country	Germany has the highest churn ($\approx 32\%$).
Gender	Slightly higher among females — needs significance testing.
Age	Older customers (40+) show more attrition.
Activity	Inactive customers churn roughly 2–3 $\times$ more than active ones.
Product holding	3–4-product customers churn most — counterintuitive; verify sample size first.

16. Business Questions — Part 2

16.1 Top 10 customers with highest churn risk

A weighted risk score was applied to retained customers to identify those most at risk of leaving.

```
/* weights: credit_score 0.4, balance 0.3, tenure 0.2, active 0.1 */
WITH churn_score AS (
  SELECT bc."customer_id", ci."SURNAME", bc."country_name", bc."age",
    ROUND(
      (0.4 * (1 - bc."credit_score" / 1000.0)) +
      (0.3 * (bc."balance" / (SELECT MAX("balance") FROM bank_churn_cleaned))) +
      (0.2 * (bc."tenure" / (SELECT MAX("tenure") FROM bank_churn_cleaned))) +
      (0.1 * (CASE WHEN bc."active_member" = 'Active Member' THEN 0 ELSE 1 END))
    , 2) AS churn_risk_score
  FROM BANK_CHURN_CLEANED bc
  JOIN CUSTOMERINFO ci ON bc."customer_id" = ci."CUSTOMERID"
  WHERE bc."churned" = 'Retain'
),
ranked AS (
  SELECT *, DENSE_RANK() OVER (ORDER BY churn_risk_score DESC) AS rank
  FROM churn_score
)
SELECT * FROM ranked WHERE rank <= 10 ORDER BY rank ASC;
```

Caveat: these weights were assigned by judgement, not fitted to data. The ranking is indicative only and should be replaced by a trained classification model before it drives customer-facing action.

16.2 Churn trend by year

Year	Total customers	Churned	Churn rate %
2019	3,302	655	19.84
2018	2,591	523	20.19
2017	2,137	478	22.37
2016	1,965	381	19.39
2015	5	0	0.00

The 2015 row contains only five customers and is excluded from trend interpretation.

16.3 Countries ranked by churn rate and credit score

Country	Avg credit score	Churn rate %	Rank
Germany	651.45	32.44	1
Spain	651.33	16.67	2
France	649.67	16.15	3

Credit scores are essentially identical across the three countries, so Germany's elevated churn cannot be attributed to creditworthiness.

16.4 Gender-based churn per country

Country	Gender	Total churn	Churn rate %
Germany	Female	448	37.55
Germany	Male	366	27.81
Spain	Female	231	21.21
Spain	Male	182	13.11
France	Female	460	20.34
France	Male	350	12.71

16.5 Correlation between age and balance (churned)

Result: -0.023 — no meaningful relationship between age and balance among churned customers.

16.6 Average salary and balance by product count

Products	Status	Avg salary	Avg balance	Customers
1	Exit	100,639.83	92,028.82	1,409
1	Retain	99,045.36	101,052.82	3,675
2	Exit	100,883.31	90,252.36	348
2	Retain	100,416.63	48,731.13	4,242
3	Exit	106,776.61	85,853.09	220
3	Retain	92,560.21	25,744.26	46
4	Exit	104,763.72	93,733.14	60

The customer counts in this table are the basis for the small-sample caveat applied throughout: **the 3- and 4-product groups hold 46 to 220 customers against 4,242 for 2-product retained.** Percentages drawn from those groups are volatile.

16.7 Consolidated insights from Part 2

Area	Insight
Top 10 high-risk	Inactive customers with low credit scores (<560) and high balances (≈180K+), concentrated in Germany and France.
Churn trend	Peaked in 2017 (22.4%), declining to 19–20% since 2018 — retention is improving.
Country ranking	Germany highest at 32.4%; credit score flat across countries, so churn is likely driven by service or competition.
Gender by country	Female churn higher in all three countries, most notably Germany (37.6%) — confirm statistically first.
Loyal customers	Mostly France-based with high estimated salaries — strong retention among financially stable long-term clients.
Age & balance	Correlation –0.023: no meaningful relationship among churned customers.
Product count	3–4-product customers churn most despite higher balances — small samples; treat as hypothesis.

17. Power BI Development

17.1 Snowflake connection

As a report developer, I want to connect Power BI to Snowflake so that I can visualise live, cleaned data.

Parameter	Value
Warehouse	BANK_ANALYTICS_WH
Database	BANK_ANALYTICS_DB
Schema	ANALYTICS
Dimensions	6 DIM_ tables (Import mode)
Fact table	FACT_CUSTOMER_CHURN_V1 (DirectQuery)
Authentication	Username & password, or SSO

```
let
    Source = Snowflake.Databases("<account>.snowflakecomputing.com", "BANK_ANALYTICS_WH"),
    DB = Source{[Name="BANK_ANALYTICS_DB", Kind="Database"]}[Data],
    ANALYTICS = DB{[Name="ANALYTICS", Kind="Schema"]}[Data],
    FACT = ANALYTICS{[Name="FACT_CUSTOMER_CHURN_V1", Kind="Table"]}[Data]
in
    FACT
```

Validation	Expected	Status
Row count	10,000	Passed
Column count	13	Passed
Data type consistency	Numeric / date / binary / varchar	Passed
Null / blank	No tolerance	Pending reverification
Sample records	Match Snowflake preview	Passed

17.2 Data enrichment

As a data analyst, I want to create derived columns so I can analyse customer behaviour in meaningful categories.

Feature	Type	Logic	Purpose
age_group	Calculated column	<30 Young; 30–40 Mid; 41–50 Mature; >50 Senior	Age segmentation
tenure_group	Calculated column	≤3 New; 4–6 Mid term; >6 Long term	Loyalty segmentation
balance_flag	Calculated column	Percentile bins (has / no balance)	Financial segmentation
score_band	Dimension table	<600 Poor; 600–700 Fair; 701–800 Good; >800 Excellent	Credit segmentation
churn_reason	Dimension table	Simulated — see Part II disclosure	Driver reporting

17.3 Key DAX measures

Measure	Formula	Description
Age Group	SWITCH(TRUE(), [Age]<26, "18–25", [Age]<36, "26–35", [Age]<46, "36–45", [Age]<56, "46–55", "56+")	Age bins
Churn Count	CALCULATE(COUNTROWS(FACT_CUSTOMER_CHURN_V1), DIM_EXIT_CUSTOMER[ExitCategory]="Exit")	Counts churned customers
Churn Rate	DIVIDE([Churn Count],[Total Customers])	Context-specific churn rate
Active %	DIVIDE([Active Members],[Total Customers])	Share of active members
Credit Card %	DIVIDE([Credit Card Holders],[Total Customers])	Cardholder penetration

Part II — Dashboard Delivery & Final Report

This part records the delivered Power BI dashboard, finalises the KPI baseline, and closes the project with limitations and recommended next steps.

18. Project Status & Scope Delivered

All six sprints defined in the BRD are complete. The pipeline runs end-to-end from the source CSV through Snowflake to a four-page Power BI report. This section records what was delivered against what was planned.

18.1 Deliverables against plan

Deliverable	Epic	Status	Note
BRD & requirements	Epic 1	Complete	v1.1, superseded by this document
Snowflake DB & schemas	Epic 2	Complete	RAW / CLEAN / ANALYTICS
Cleaning & transformation	Epic 2	Complete	Python + SQL, 10,000 rows validated
Star-schema model	Epic 3	Complete	1 fact + 6 dimensions
EDA report	Epic 3	Complete	Univariate, bivariate, correlation
Power BI dashboard	Epic 4	Complete	4 pages — see Section 20
Row-level security	Epic 5	Not implemented	Deferred — see Section 22
Scheduled refresh pipeline	Epic 5	Not implemented	Deferred — see Section 22
Documentation package	Epic 6	Complete	This document

18.2 Revised KPI baseline

The BRD set an initial churn target of $\leq 10\%$. That figure was aspirational rather than evidence-based, and against an observed rate of 20.37% it produced a gap too wide to plan against. The delivered dashboard therefore standardises on a **16.5% target**, set at the level of the best-performing country cohort (France, 16.15%) rounded up. This makes the target internally achievable — it asks the business to bring Germany and Spain toward what France already demonstrates is possible — rather than benchmarking against an external figure the data cannot support.

All targets in this document, and every reference on the dashboard, use 16.5%. Where earlier drafts cite 10% or 15%, this section supersedes them.

18.3 Final KPI results

KPI	Result	Target	Status
Overall churn rate	20.37%	≤ 16.5%	Above target
Retention rate	79.63%	≥ 83.5%	Below target
Average tenure	4.86 yrs	—	Baseline
Active member ratio	52%	≥ 70%	Below target
Credit card penetration	70.66%	—	Baseline
Total customers exited	2,037	—	Baseline
Highest-risk region	Germany 32.44%	≤ 16.5%	Priority
Highest-risk product	Mortgage 100%	—	Investigate

19. Dashboard Architecture

The report is built as a four-page narrative rather than a single dense canvas. A navigation landing page routes users to three analytical pages, each answering a different class of question. Every page carries a persistent left-hand summary panel with a Key Insights tooltip, so a reader arriving cold can orient without a briefing.

19.1 Page structure

Page	Audience	Question it answers
Report Overview	All users	Where do I go for what?
Executive Overview	Management	How bad is churn, where, and is it improving?
Demographic & Profiling	Retention / CRM	Who is leaving, and at what life stage?
Financial & Product	Product / Risk	Which products and credit profiles drive churn?

19.2 Design conventions

- **Colour semantics:** dark green denotes headline or negative-attention KPIs; yellow denotes retained customers; red denotes exited customers. This mapping is consistent on every page.
- **Persistent summary panel:** each page repeats period, key ratios, and an information tooltip in a fixed left column, so context travels with the user.
- **Drill-through:** the Executive page links through to Demographic and Financial detail, keeping the landing view uncluttered.
- **Source attribution:** a footer on every page names the dataset and record count, so no figure appears without provenance.
- **Model:** dimensions imported, fact table on DirectQuery — balancing refresh stability against live fact updates.

20. Dashboard Design & Layout

20.1 Report Overview (landing page)

The entry point. Three cards summarise what each analytical page contains and what its key metrics mean, with navigation buttons on the left. The Financial card additionally documents how to read the scatter plot — axis definitions and colour coding — which removes the most common source of misreading before the user arrives at that page.



Figure 13 — Report Overview: navigation and page summaries

20.2 Executive Overview

The management landing page. Four KPI cards carry the headline numbers, each annotated with a plain-language interpretation rather than a bare figure. A trend line tracks churn against the 16.5% target across all four years, a pie and table break churn down by country, and a treemap ranks the top ten churn drivers.

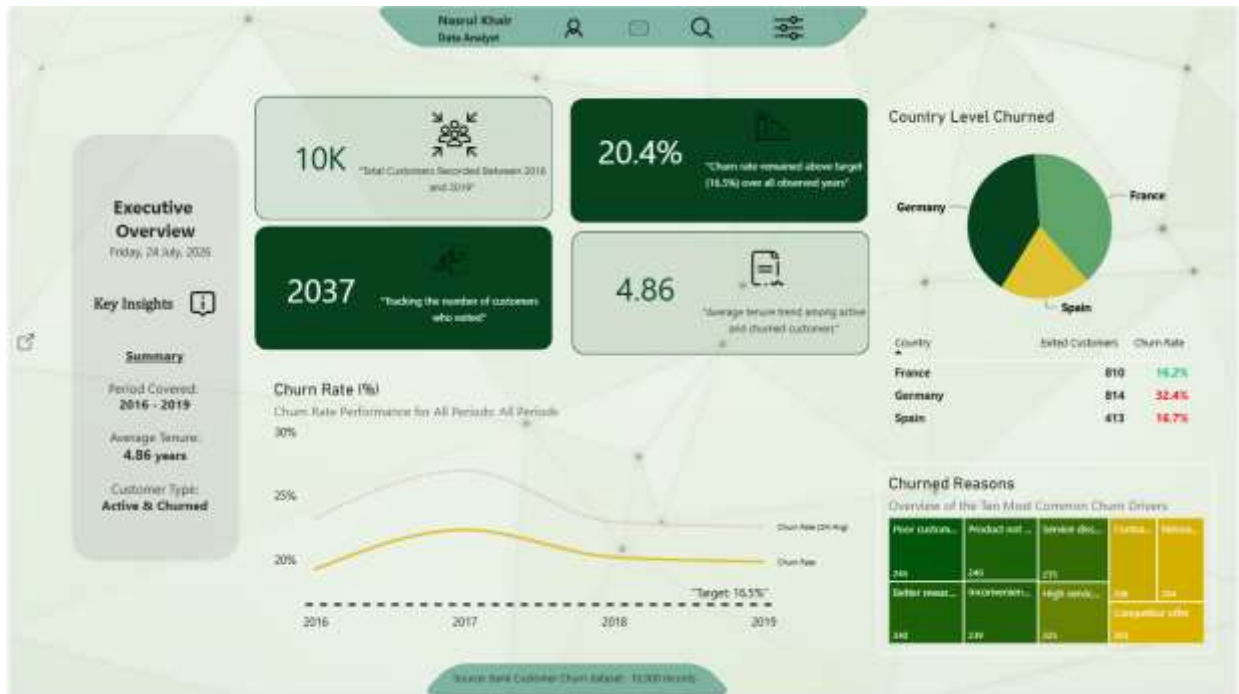


Figure 14 — Executive Overview

Key highlights

- 10,000 customers recorded between 2016 and 2019; 2,037 exited, giving a 20.37% churn rate.
- Churn stayed above the 16.5% target in every observed year — this is a structural issue, not a single bad period.
- The trend peaked at 22.37% in 2017 and eased to 19.84% by 2019 — improving, but not yet at target.

Country-level insights

- Germany 32.44%, Spain 16.67%, France 16.15% — Germany is roughly double the other two.
- France carries the largest base (5,014 customers) at the lowest churn rate, which is what makes 16.5% a demonstrably reachable target.
- Average credit score is essentially flat across all three countries (649.67–651.45), so Germany's excess churn is not a creditworthiness story.

Churn drivers

The Churned Reasons treemap displays the ten most common exit reasons. **These reason codes are simulated, not observed** — the source dataset contains no exit-survey field. They are included to demonstrate the reporting pattern and should not be cited as findings. The top categories cluster around service quality, pricing, and rewards, which is the hypothesis a genuine exit-survey dataset should be built to test.

20.3 Demographic & Profiling

Segmentation view for retention and CRM teams. Three KPI cards establish population size and engagement levels; the central bar chart plots churn volume by tenure year with a click-through to top reasons; the right column stacks gender, country, and credit-score breakdowns for rapid comparison.

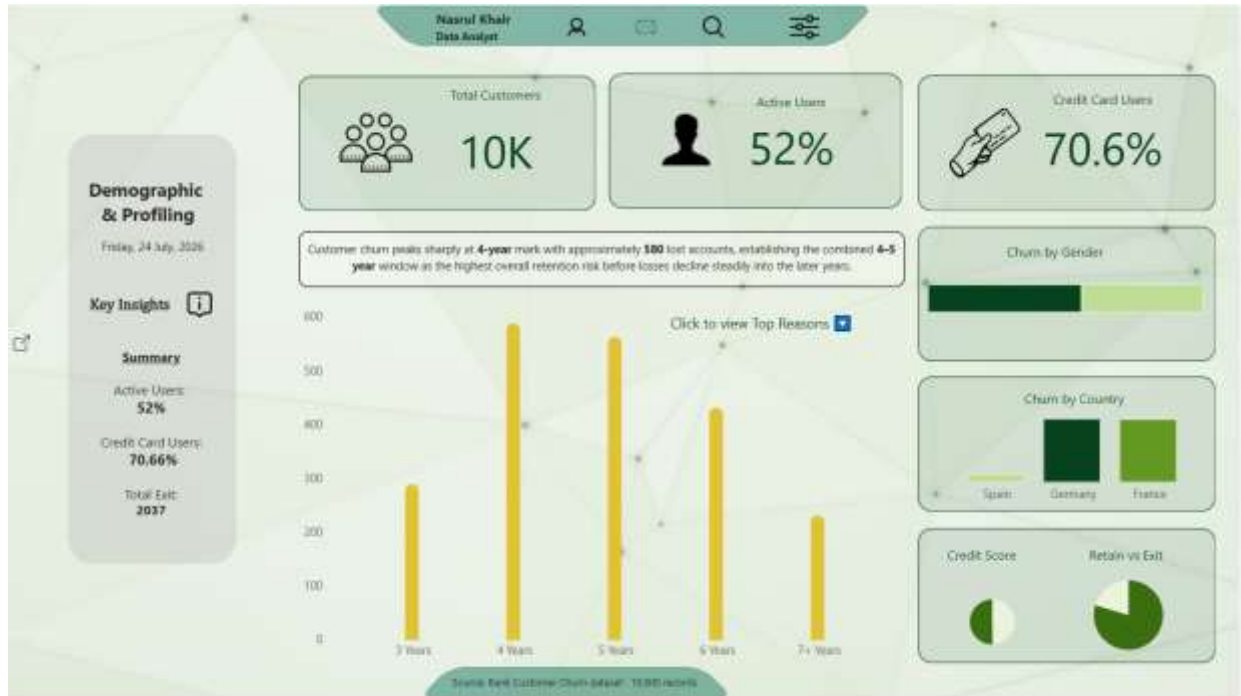


Figure 15 — Demographic & Profiling

Key insights

- **Tenure is the sharpest signal on this page.** Churn peaks at the 4-year mark with roughly 580 lost accounts, and the combined 4–5 year window is the single highest-risk retention stage before losses decline into later years.
- **Activity status:** only 52% of customers are active. Inactive members churn roughly 2–3× more than active ones — the clearest and most consistent behavioural signal in the dataset.
- **Age:** churn rises steeply with age, from 7.56% under 30 to 56.04% in the 50s bracket. Correlation with churn is +0.29, the strongest single numeric relationship present.
- **Gender:** female churn (25.07%) exceeds male (16.46%), most pronounced in Germany at 37.55%. Correlation is weak (0.11); treat as a hypothesis pending a significance test.
- **Credit card ownership:** 70.66% penetration, with non-cardholders churning more — a visible cross-sell opportunity.

Narrative summary

Churn concentrates among older, inactive customers at the 4–5 year tenure mark. Behavioural engagement is the most defensible lever available: it has the strongest evidence base and is directly actionable through campaign design. Demographic effects such as gender and geography are weaker and should be validated statistically before campaigns are built around them.

20.4 Financial & Product

Product and risk view. Three KPI cards headline overall churn, average churned balance, and the highest-risk product segment. Below, a credit-score band comparison and a category-versus-churn bar sit alongside a scatter plot of credit score against account balance on a log scale, with retained and exited customers colour-separated. The right column holds the product segment matrix and a lock-in tier distribution.



Figure 16 — Financial & Product

Key insights

- **Product segment is the strongest differentiator on this page.** Churn ranges from 7.6% (Savings) to 100% (Mortgage) — a spread far wider than any demographic variable produces.
- **Average churned balance of 91K exceeds the retained average,** confirming that the bank is losing financially significant customers rather than marginal ones. This raises the revenue stakes of the 20.37% rate considerably.
- **Credit score is a weak discriminator:** the Poor-to-Excellent band spread runs only about 20% to 25%, and the correlation with churn is -0.03 . The scatter plot shows red and yellow points thoroughly intermixed across the full 400–900 range, which is the visual confirmation.
- **Lock-in tier:** Low and Medium lock-in tiers hold the overwhelming majority of customers, while High and Extreme tiers are sparse — indicating the portfolio has limited structural retention built into its product mix.

Product segment matrix

Product category	Churn rate	Read
Savings	7.6%	Strongest retention — the benchmark segment
Transactional	27.7%	Above overall rate; largest customer volume
Credit Card	82.7%	Severe — but small base, verify before acting
Mortgage	100.0%	Every customer exited — see caveat below

Caveat: the Mortgage (100%) and Credit Card (82.7%) figures rest on small sample sizes — 46 to 220 customers, against 4,242 for the two-product Savings group. A 100% rate on a small base is far more likely to reflect sample scarcity than a genuine total-loss segment. These figures should be treated as flags for investigation, not as validated findings, and the dashboard should carry a visible sample-size indicator on this matrix before the numbers are presented to stakeholders.

21. Consolidated Recommendations

Recommendations are ordered by strength of evidence rather than by size of the headline number. The first three rest on consistent, well-sampled signals; the remainder require validation before resourcing.

21.1 Priority actions

#	Action	Evidence base	Confidence
1	Re-engagement programme for inactive members	48% of base inactive; 2–3× churn multiple; consistent across every cut	High
2	Retention intervention at the 3–4 year tenure mark	Clear volume peak at year 4 (~580 accounts) visible in tenure distribution	High
3	Germany market investigation — service and competitor review	32.44% vs 16.15% France, with credit score flat across countries	High
4	Loyalty benefits for customers aged 45+	Age correlation +0.29; 56.04% churn in 50s bracket	Medium
5	Cross-sell savings products to single-product holders	Savings segment churns at 7.6% vs 27.7% transactional	Medium
6	Review mortgage and credit-card servicing	100% / 82.7% churn — but small samples	Low — verify first
7	Gender-targeted engagement in Germany	37.55% female churn, correlation only 0.11	Low — test first

21.2 Why engagement leads

Of every variable examined, activity status combines the three properties that matter for action: it is strongly and consistently associated with churn, it applies to nearly half the customer base, and it is directly modifiable through campaign activity. Age and geography are descriptive — the bank cannot change a customer's age or country. Product mix is modifiable but rests on thin samples. Engagement is both well-evidenced and actionable, which is why it heads the list despite Germany and Mortgage carrying larger headline percentages.

22. Limitations & Data Quality Notes

Recording these plainly is part of the deliverable. Each item below constrains how far the findings above can be pushed.

22.1 Data limitations

Limitation	Impact	Mitigation
Simulated churn reason codes	Exit reasons cannot be cited as findings	Commission an exit-survey dataset
Small samples in 3–4 product segments	Mortgage / Credit Card rates unreliable	Add sample-size indicator; collect more data
Null verification incomplete in RAW layer	Residual risk of undetected gaps	Re-run validation before production sign-off
Static extract, 2016–2019	No live view of current performance	Implement scheduled refresh (Epic 5)
No customer-value field	Churn measured by count, not revenue impact	Join to revenue data for weighted churn
Single-source dataset	No behavioural or interaction history	Integrate CRM and transaction logs

22.2 Analytical limitations

- Correlations are weak across the board — the strongest is age at +0.29. No single variable explains churn; the findings are directional, not predictive.
- The Top-10 risk ranking uses manually assigned weights (0.4 credit score, 0.3 balance, 0.2 tenure, 0.1 activity). These were chosen by judgement, not fitted to data, and a trained classification model should replace them before the ranking drives any customer-facing action.
- Gender and geography differences have not been significance-tested. They are reported as observed patterns pending a chi-square test.
- Correlation does not establish causation anywhere in this analysis — particularly for the inactivity finding, where disengagement may equally be an early symptom of intent to leave rather than its cause.

22.3 Scope deferred

Row-level security (Epic 5) and the scheduled refresh pipeline (Epic 5) were specified in the BRD but not implemented. Both require a Power BI Pro licence and gateway infrastructure outside the scope of this build. They are prerequisites for production deployment and are carried into the next phase.

23. Next Steps

Phase	Action	Prerequisite
Immediate	Add sample-size indicators to the product segment matrix	None
Immediate	Re-run null validation across the RAW layer	Snowflake access
Immediate	Significance-test the gender and geography differences	None
Short term	Implement row-level security by geography	Power BI Pro licence
Short term	Configure scheduled refresh via gateway	Gateway infrastructure
Short term	Replace manual risk weights with a fitted classification model	Cleaned training set
Medium term	Source an exit-survey dataset for genuine reason codes	Business sponsorship
Medium term	Join customer revenue data for value-weighted churn	Revenue data access
Medium term	Extend the series beyond 2019 for current-state monitoring	Updated extract

24. Conclusion

The project delivered a complete analytical pipeline — from an unstructured CSV through a layered Snowflake warehouse and star-schema model to a four-page Power BI report — against a 20.37% churn rate that has remained above target in every observed year.

The most defensible finding is behavioural rather than demographic: customer engagement, not credit profile, geography, or age, is the lever with both the strongest evidence and the clearest path to action. Germany and the mortgage segment carry larger headline percentages, but the first requires investigation to explain and the second rests on samples too small to trust. Re-engaging the 48% of the base that is currently inactive is where the analysis points most confidently.

Equally, the analysis establishes what this dataset cannot answer. Correlations are weak throughout, exit reasons are simulated, and the series ends in 2019. The dashboard is a monitoring instrument and a hypothesis generator — the next phase of value comes from better data feeding it, not from further analysis of the same extract.

End of report — Customer Churn Analysis & Retention Strategy v2.0 (Final)